

Brian Velazquez

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EDUCATION

Florida Gulf Coast University

Bachelor of Science in Software Engineering

GPA: 4.0

Relevant Coursework: *Data Structures & Algorithms, Data Engineering, Software Testing, Requirements Eng. & Analysis*

Fort Myers, FL

Expected May 2027

PROJECTS

GridWatch - Drone Route Optimizer

October 2025

- Built optimization pipeline with Python and Google OR-Tools to generate efficient drone routes for power grid inspection while respecting battery constraints
- Implemented route expansion algorithm using predecessor matrices and Dijkstra pathfinding to reconstruct full flight paths from optimized waypoints
- Developed React dashboard with Plotly.js and Mapbox integration featuring mission animation, battery monitoring, and real-time route visualization
- Designed Flask API backend to serve optimized mission data with caching system reducing recalculation overhead

Full-Stack To-do Application

July 2025

- Built REST API backend with C++ Crow framework and PostgreSQL database implementing CRUD operations and session-based authentication
- Implemented user authentication with password hashing and HTTP cookie session management with automatic expiration
- Designed database schema with foreign key constraints and prepared statements to prevent SQL injection and ensure data integrity
- Built SPA frontend using vanilla JavaScript and modal task editing interface

Wayfinder

September 2025

- Developed cross-platform mobile application using React Native and Expo with Firebase authentication for secure user registration and session management
- Integrated Google Places API, Google Vision API for AI facial recognition, and React Native Maps for location-based challenges and badge collection

Zonta Club of Naples Website

September 2025 - December 2025

- Developed a REST API using Node.js and Express, integrating Stripe webhooks for secure donation processing and automated membership subscription management
- Implemented automated background schedulers with node-cron to trigger targeted email notifications for event reminders and membership renewals via Nodemailer
- Designed MongoDB pipelines to generate real-time finance statistics and utilized Cloudinary with Multer for cloud-based document storage

EXPERIENCE

IT Intern

January 2026 - Present

Bonita Springs Utilities

Bonita Springs, FL

- Developed and executed SQL queries to generate database extracts and detailed reports, optimizing data retrieval processes
- Analyzed and troubleshooted off-the-shelf applications to identify problem areas and resolve technical issues
- Configured and maintained hardware and software infrastructure while resolving technical user issues

AWARDS

Microsoft Office Specialist Florida State Champion

December 2022

KnightHacks VIII: NextEra Energy Drone Optimization Challenge Winner

October 2025

CAMPUS & COMMUNITY INVOLVEMENT

Software Engineering Club: *Member*

August 2025 - Present

ShellHacks 2025: *Participant*

September 2025

KnightHacks VIII 2025: *Participant*

October 2025

TECHNICAL SKILLS

Languages: C/C++, JavaScript, Python

Certifications: Microsoft Office Specialist, CompTIA CSIS, Cisco CSST Cybersecurity, IT Specialist - Cybersecurity

Frameworks & Libraries: React, React Native, Node.js, Express.js, CrowCpp, Qt

Operating Systems: Linux, Windows

Developer Tools: GitHub, VS Code, Visual Studio, CMake, vcpkg

Entry Level Knowledge: PostgreSQL, Firebase, MongoDB, Cloudinary